Base employés

1. SQL :
```
select *
from dept
```

```
AR : res = dept
```

Sur une requête select * from nomtable, je ne peux pas avoir d'erreur, c'est à partir du moment où je rajoute d'autres clauses que je dois faire des vérifications.

2. SQL :
```
select * from dept
where deptno > 10
```

```
AR : res =  restrict(emp, deptno>10)
```

Résultat attendu :
`select * from dept;` → me donne la liste des départements, sur cette requête pas de risque d'erreur

Contenu de `dept`

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Parmi ces départements, je veux ceux de numéro supérieur à 10, il doit donc me rester dans mon résultat

DEPTNO	DNAME	LOC
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

3. SQL :
```
select *
from dept
order by loc
```

```
AR : pas d'ordre en algèbre relationnelle
```

Résultat attendu :
`select * from dept;` → me donne la liste des départements

Contenu de `dept`

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Mon résultat doit correspondre à ces données triées par localisation, soit

DEPTNO	DNAME	LOC
40	OPERATIONS	BOSTON
30	SALES	CHICAGO
20	RESEARCH	DALLAS
10	ACCOUNTING	NEW YORK

4. SQL :
```
select ename, mgr
from emp
```

AR : `res = proj(emp, {ename, mgr})`

Résultat attendu :
`select * from emp;` → me donne la liste des employés

Contenu de `emp`

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
9999	GAG				3000		10
7839	KING	PRESIDENT		17/11/81	5000		10
7698	BLAKE	MANAGER	7839	21/01/81	2850		30
7782	CLARK	MANAGER	7839	09/06/81	2450		10
7566	JONES	MANAGER	7839	02/04/81	2975		20
7654	MARTIN	SALESMAN	7698	28/09/81	1250	1400	30
7499	ALLEN	SALESMAN	7698	20/02/81	1600	300	30
7844	TURNER	SALESMAN	7698	08/09/81	1500	0	30
7900	JAMES	CLERK	7698	03/12/81	950		30
7521	WARD	SALESMAN	7698	22/02/81	1250	500	30
7902	FORD	ANALYST	7566	03/12/81	3000		20
7369	SMITH	CLERK	7902	17/12/80	800		20
7788	SCOTT	ANALYST	7566	31/08/82	3000		20
7876	ADAMS	CLERK	7788	04/10/82	1100		20
7934	MILLER	CLERK	7782	23/01/82	1300		10

Mon résultat doit conserver uniquement les noms et numéros de manager des employés, soit les colonnes ename et mgr, soit

ENAME	MGR
GAG	
KING	
BLAKE	7839
CLARK	7839
JONES	7839
MARTIN	7698
ALLEN	7698
TURNER	7698
JAMES	7698
WARD	7698
FORD	7566
SMITH	7902
SCOTT	7566
ADAMS	7788
MILLER	7782

5.
```
select *
from emp
where job='SALESMAN'
```

6.
```
select ename, job, sal*12
from emp
where job in ('ANALYST','MANAGER')
```

7.
```
select empno, ename
from emp
where mgr is NULL
```

8. SQL :
```
select empno, ename
from emp
where empno between 7800 and 7899
```

AR :
```
r1 = restrict(emp, empno>=7800 ∧ empno<=7899)
res = proj(r1, {empno, ename})
```

9.
```
select empno, ename
from emp
where ename like 'A%'
```

10.
```
select ename, job, round(sal*1.82,2)
  from emp
```

11. SQL :
```
  select distinct job
  from emp
```

  AR : r1 = proj(emp, {job})

12.
```
  select *
  from emp
  where comm > sal
```

13.
```
  select *
  from dept
  where deptno > 20
```

14.
```
  select *
  from emp
  where job in ('ANALYST','CLERK')
```

15.
```
  select *
  from emp
  where ename like '_L%'
```

16.
```
  select *
  from emp
  where comm is NULL
```

17.
```
  select *
  from emp
  order by job, sal desc
```

18.
```
  select *
  from emp
  where job in ('ANALYST','CLERK')
    and deptno = 10
```

19.
```
  select ename, sal
  from emp
  where sal > 3000
```

20.
```
select ename "Nom", sal "Salaire" from emp
```

21.
```
  select ename, greatest(sal,comm)
  from emp
  where comm is NOT NULL
```

22.
```
  select *
  from emp
  order by deptno desc, sal
```

23.
```
  select ename, floor(sal*1.32)
  from emp
  where deptno = 20
```

24.
```
  select mgr
  from emp
  where deptno in (10,20,30)
```

```
25.select *
   from emp
   where (ename like '%J%' OR ename like '%K%')
     and job != 'MANAGER'
     order by ename, deptno

26.select *
   from dept
   where dname like '%L%L%'

27.select sal, job
   from emp
   order by job, sal desc

28.select empno, ename, job
   from emp
   where deptno not in (10,20,30)

29.select distinct deptno
   from emp

30.select ename, sal, job
   from emp
   where job not in ('ANALYST','MANAGER','CLERK')
     and sal > 12000
```

Base vols

```
31.select *
   from iin_vols

32.select *
   from iin_avions
   where capacite > 300

33.select *
   from iin_vols
   where depart='Paris'

34.select nom, habite
   from iin_pilotes

35.select *
   from iin_vols
   where depart='Paris'
      or arrivee='Paris'

36.select nom
   from iin_pilotes
   where nom like 'S%'

37.select distinct type
   from iin_avions
```

```
38.select *
   from iin_vols
   where pilote=1
     and copilote=2

39.select *
   from iin_vols
   where (pilote=1 and copilote=2)
      or (pilote=2 and copilote=1)

40.select navion, capacite
   from iin_avions
   where type='Airbus'
   order by capacite desc

41.select *
   from iin_vols
   where depart='Paris'
     and arrivee!='Nice'

42.select nvol
   from iin_vols
   where depart='Paris'
     and arrivee!='Nice'

43.select nvol, depart, arrivee
   from iin_vols
   where depart='Paris'
      or depart='Nice'

   ou

   select nvol, depart, arrivee
   from iin_vols
   where depart in ('Paris', 'Nice')

44.select navion
   from iin_avions
   where capacite >=200
     and capacite <= 600

   ou

   select navion
   from iin_avions
   where capacite between 200 and 600
```

Base gaulois

```
45.select *
   from iin_gaulois
   where nom = 'Asterix'

46.select distinct metier
   from iin_gaulois
   where sexe = 'F'
```

```
47.select *
   from iin_gaulois
   where sexe = 'F'
     and metier is NULL

48.select *
   from iin_gaulois
   where metier = 'guerrier'
     and vilno = 8

49.select *
   from iin_gaulois
   where vilno between 5 and 10
   order by vilno, metier

50.select *
   from iin_bataille
   where nom is not NULL
   order by nom

51.select *
   from iin_bataille
   order by datebat
```

Base inventions

```
52.select nation, population/area
   from nation
```

Base commerciale

```
53.select *
   from product
   order by descrip

54.select *
   from item
   where ordid=610
   order by itemid

55.select ordid, orderdate, total
   from ord where custid=100
   order by orderdate
```